FUNCTION	EQUATION FOR CALCULATING
Secant (X) =	1/COS(X)
Cosecant (X) =	1/SIN(X)
Cotangent (X) =	1/TAN(X)
Arc Sine (X) =	ATN(X/SQR(1-X*X))
Arc Cosine (X) =	-ATN(X/SQR(1-X*X)) + PI/2
Arc Secant (X) =	ATN(SQR(X*X-1)) + (SGN(X)-1) * PI/2
Arc Cosecant (X) =	ATN(1/SQR(X*X-1)) + (SGN(X)-1) * PI/2
Arc Cotangent (X) =	-ATN(X) + PI/2
Hyperbolic Sine (X) =	(EXP(X)-EXP(-X))/2
Hyperbolic Cosine (X) =	(EXP(X)+EXP(-X))/2
Hyperbolic Tangent (X) =	(EXP(X)-EXP(-X))/(EXP(X)+EXP(-X))
Hyperbolic Secant (X) =	1/HYPERBOLIC COSINE
Hyperbolic Cosecant (X) =	1/HYPERBOLIC SINE
Hyperbolic Cotangent (X) =	1/HYPERBOLIC TANGENT
Arc Hyperbolic Sine (X) =	LOG(X+SQR(X*X+1))
Arc Hyperbolic Cosine (X) =	LOG(X+SQR(X*X-1))
Arc Hyperbolic Tangent (X) =	LOG((1+X) /(1-X))/2
Arc Hyperbolic Secant (X) =	LOG((SQR(1-X*X)+1)/X)
Arc Hyperbolic Cosecant (X) =	LOG(1+SGN(X)*SQR(1+X*X))/X
Arc Hyperbolic Cotangent (X) =	LOG((X+1)/(X-1))/2
$LOG_{10}(X)$ =	LOG(X)/LOG(10) = 0.4342945 LOG(X)
PI =	4*ATN(1) = 3.141592653589794

TABLE 4-7
Calculating rarely used functions from more common ones. All angles are in radians, ATN(X) is the arctangent, LOG(X) is the natural logarithm, SGN(X) is the sign of X (i.e., -1 for $X \leq 0$, 1 for $X>0$), EXP(X) is e^X.

Another option is to precalculate these slow functions, and store the values in a **look-up table (LUT)**. For example, imagine an 8 bit data acquisition system used to continually monitor the *voltage* across a resistor. If the parameter of interest is the *power* being dissipated in the resistor, the measured voltage can be used to calculate: $P = V^2/R$. As a faster alternative, the power corresponding to each of the possible 256 voltage measurements can be calculated beforehand, and stored in a LUT. When the system is running, the measured voltage, a digital number between 0 and 255, becomes an index in the LUT to find the corresponding power. Look-up tables can be hundreds of times faster than direct calculation.

Third, learn what is *fast* and what is *slow* on your particular system. This comes with experience and testing, and there will always be surprises. Pay particular attention to *graphics* commands and *I/O*. There are usually several ways to handle these requirements, and the speeds can be tremendously different. For example, the BASIC command: *BLOAD*, transfers a data file directly into a section of memory. Reading the same file into memory byte-by-byte (in a loop) can be 100 times as slow. As another example, the BASIC command: *LINE*, can be used to draw a colored box on the video screen. Drawing the same box pixel-by-pixel can also take 100 times as long. Even putting a *print* statement within a loop (to keep track of what it is doing) can slow the operation by *thousands*!